Cryosphere Geophysics And Remote Sensing (CryoGARS) group
Department of Geosciences and CGISS, Boise State University
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Relevant Experience

HP has a background in both engineering and science; 28 yrs as a snow scientist, and 20 yrs developing FMCW radar systems. He has successfully led 27 research grants from federal and state agencies. He has supported SWE remote sensing efforts in Europe (ESA CoreH2O) as well as the U.S. (CLPX I/II/III), and was the Project Scientist for the NASA SnowEx Mission from 2018-2021, and on the leadership team from 2016-2023.

Education

2005	Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0 Institute of Arctic and Alpine Research, University of Colorado at Boulder Thesis: *Snowpack spatial variability: towards understanding its effect on remote sensing measurements and snow slope stability*
1999	B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0 University of Washington, Thesis: *Wet snow densification*

Selected Professional Experience

08/2018-8/2021	Project Scientist, NASA SnowEx Mission
08/2016-07/2018	Co-lead of Field Campaign, NASA SnowEx 2017 Mission
08/2022-present	Professor, Department of Geosciences, Boise State University

Selected Honors and Awards

2020	BSU Faculty Excellence Award
2019	NASA Instrument Incubator Program Award, NASA, 1st in Idaho history
2019	2018 WRR Editors' Choice Award: Hedrick et al., 2018
2018	NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
2012	Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
2010-13	New Investigator Program Award, NASA, 1st in Idaho history
2010	Young Scientist Award, Cryosphere Section, American Geophysical Union
2009	Top Reviewer Award, Cold Regions Science and Technology (CRST)
2008	U.S. Young Scientist Award, International Union of Radio Science (URSI)

Selected Relevant Peer-Reviewed Publications

[total=86, h-index=33, i10-index=70, citations=3306]

2024

Oveisgharan, S., Zinke, R., Hoppinen, Z., and **Marshall, H.P.** (2024). Snow Water Equivalent Retrieval Over Idaho, Part A: Using Sentinel-1 Repeat-Pass Interferometry. *The Cryosphere*, 18:559–574, doi.org/10.5194/tc–18–559–2024

Hoppinen, Z. M., Oveisgharan, S., **Marshall, H.P.**, Mower, R., Elder, K., and Vuyovich, C. (2024). Snow Water Equivalent Retrieval Over Idaho, Part B: Using L-band UAVSAR Repeat-Pass Interferometry. *The Cryosphere*, 18:575–592, doi.org/10.5194/tc–18–575–2024

2023 Bonnell, R., McGrath, D., Hedrick, A. R., Trujillo, E., Meehan, T. G., Williams, K., **Marshall, H.P.**, Sexstone, G., Fulton, J., Ronayne, M. J., Fassnacht, S. R., Webb, R. W., and Hale, K. E. (2023). Snowpack relative permittivity and density derived from near-coincident lidar and ground-penetrating radar. *Hydrological Processes*, 37(10):e14996

Tarricone, J., Webb, R., **Marshall, H.P.**, Nolin, A., and Meyer, F. (2023). Estimating snow accumulation and ablation with l-band insar. *The Cryosphere*, 17(5):1997–2019, doi.org/10.5194/tc–17–1997–2023

2022 Lievens, H., Brangers, I., **Marshall, H.P.**, Jonas, T., Olefs, M., and DeLannoy, G. (2022). Sentinel-1 snow depth retrieval at sub-kilometer resolution over the European Alps. *The Cryosphere*, 16(1):159–177, doi.org/10.5194/tc–16–159–2022

Lund, J., Forster, R., Deeb, E., Liston, G., Skiles, M., and **Marshall, H.P.** (2022). Interpreting sentinel-1 sar backscatter signals of snowpack surface melt/freeze, warming, and ripening, through field measurements and physically-based snowmodel. *Remote Sensing*, 14:4002, https://doi.org/10.3390/rs14164002

McGrath, D., Bonnell, R., Zeller, L., Olsen-Mikitowicz, A., Bump, E., Webb, R., and **Marshall, H.P.** (2022). A time-series of snow density and snow water equivalent observations derived from the integration of gpr and uav sfm observations. *Frontiers in Remote Sensing*, 3:886747, doi:10.3389/frsen.2022.886747

Tsang, L., Durand, M., Derksen, C., Barros, A. P., Kang, D.-H., Lievens, H., **Marshall, H.P.**, Zhu, J., Johnson, J., King, J., Lemmetyinen, J., Sandells, M., Rutter, N., Siqueira, P., Nolin, A., Osmanoglu, B., Vuyovich, C., Kim, E. J., Taylor, D., Merkouriadi, I., Brucker, L., Navari, M., Dumont, M., Kelly, R., Kim, R. S., Liao, T.-H., and Xu, X. (2022). Review article: Global monitoring of snow water equivalent using high frequency radar remote sensing. *The Cryosphere*, 16(9):3531–3573

2021 **Marshall, H.P.**, Deeb, E., Forster, R., Vuyovich, C., Elder, K., Hiemstra, C., and Lund, J. (2021). L-band InSAR depth retrieval during the NASA SnowEx 2020 Campaign: Grand Mesa, Colorado. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*, pages 625–627, DOI:10.1109/IGARSS47720.2021.9553852

2019 Lievens, H., Demuzere, M., **Marshall, H.P.**, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Girotto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Schöber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. *Nature Communications*, 10(1):1–12

McGrath, D., Webb, R., Shean, D., Bonnell, R., **Marshall, H.P.**, Painter, T., Molotch, N., Elder, K., Hiemstra, C., and Brucker, L. (2019). Spatially extensive ground-penetrating radar snow depth observations during NASA's 2017 SnowEx Campaign: Comparison to in situ, airborne, and satellite observations. *Water Resources Research*, 55(11):10026–10036, doi:10.1029/2019WR024907

2008 **Marshall, H.P.** and Koh, G. (2008). FMCW radars for snow research. *Cold Regions Science and Technology*, 52(2):118–131

2007 **Marshall, H.P.**, Schneebeli, M., and Koh, G. (2007). Snow stratigraphy measurements with high-frequency FMCW radar: Comparison with snow micro-penetrometer. *Cold Regions Science and Technology*, 47(1-2):108–117